A Project report on

UNIFIED ENSEMBLE LEARNING FOR CYBER ATTACK DETECTION AND CLASSIFICATION IN WIRELESS SENSOR NETWORKS

Submitted in partial fulfillment of the requirements

for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

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Rotarypuram Village, B K Samudram Mandal, Ananthapuramu - 515701

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COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

Certificate

This is to certify that the project report entitled Unified Ensemble Learning for Cyber Attack Detection and Classification in Wireless Sensor Networks is the bonafide work carried out by **B. Lakshmi Moksha** bearing Roll Number **22G1A3246** in partial fulfilment of the requirements for the award of the degree of **Computer Science and Engineering (Data Science)** during the academic year 2025-2026.

Project Guide

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I Ms. B. Lakshmi Moksha bearing reg no: 224G1A3246 student of **Computer Science and Engineering (Data Science)**, **SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY (AUTONOMOUS)**, Rotarypuram, hereby declare that the dissertation entitled **"UNIFIED ENSEMBLE LEARNING FOR CYBER ATTACK DETECTION AND CLASSIFICATION IN WIRELESS SENSOR NETWORKS"** embodies the report of our project work carried out by us during IV year under the guidance of Dr. P. Chitralingappa, M.Tech.,Ph.D., Associate Professor & HOD, Department of Computer Science and Engineering (Data Science), Srinivasa Ramanujan Institute of Technology, and this work has been submitted for the partial fulfillment of the requirements for the award of degree of Bachelor of Technology.

The results embodied in this project report have not been submitted to any other University or Institute for the award of any Degree or Diploma.

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ACKNOWLEDGEMENT

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Project Associate

224G1A3246

ABSTRACT

Wireless Sensor Networks (WSNs) are widely used in applications such as environmental monitoring, healthcare, and industrial automation. However, due to their open wireless communication and limited computational resources, WSNs are highly vulnerable to cyberattacks. Traditional detection systems are often inadequate in accurately classifying different types of attacks and fail to detect unknown or emerging threats in dynamic network environments.

To address these challenges, this project proposes a unified ensemble learning-based cyberattack detection and classification system for Wireless Sensor Networks. The system integrates multiple machine learning and deep learning models, including Random Forest, XGBoost, Multilayer Perceptron (MLP), and Autoencoder, to enhance detection accuracy, classification stability, and overall system reliability. Supervised learning models are used to classify known attacks such as Blackhole, Grayhole, Flooding, and TDMA attacks, while the Autoencoder is employed for anomaly detection to identify unknown or emerging threats.

The proposed model is trained using the WSN-DS dataset and evaluated using network traffic generated from the COOJA simulation environment to ensure realistic testing conditions. The system achieves a high detection accuracy of 99.88% and supports real-time monitoring, forensic logging, and prevention recommendations. This makes the proposed system efficient and suitable for real-world Wireless Sensor Network security applications.

Keywords: Cyber Security, Wireless Sensor Networks, Ensemble Techniques, Cyber Attacks, COOJA Simulator

PROJECT CONTRIBUTION TOWARDS SUSTAINABLE DEVELOPMENT GOALS

Project Title:
Unified Ensemble Learning for Cyber Attack Detection and Classification in Wireless Sensor Networks

SDG - Project Mapping:

S.No.	Sustainable Development Goals	Mapped (Yes/No)
1	SDG 1: No Poverty	No
2	SDG 2: Zero Hunger	No
3	SDG 3: Good Health and Well-Being	Yes
4	SDG 4: Quality Education	Yes
5	SDG 5: Gender Equality	No
6	SDG 6: Clean Water and Sanitation	No
7	SDG 7: Affordable and Clean Energy	Yes
8	SDG 8: Decent Work and Economic Growth	Yes
9	SDG 9: Industry Innovation and Infrastructure	Yes
10	SDG 10: Reduced Inequalities	No
11	SDG 11: Sustainable Cities and Communities	Yes
12	SDG 12: Responsible Consumption and Production	No
13	SDG 13: Climate Action	No
14	SDG 14: Life Below Water	No
15	SDG 15: Life on Land	No
16	SDG 16: Peace Justice and Strong Institutions	Yes
17	SDG 17: Partnerships for the Goals	No

Explanation:

- Enhances network security by detecting and classifying cyberattacks in Wireless Sensor Networks.
- Enables real-time monitoring and analysis of network activities to prevent cyber threats.
- Improves system reliability and data integrity by reducing attacks.

Expected Impact:

Parameter	Impact
Cyberattack Detection Accuracy	Up to 99+%
Threat Detection Speed	Real-time detection
Network Security Improvement	High
Anomaly Detection Capability	Detects unknown attacks

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LIST OF ABBREVIATIONS

AE	Autoencoder
AUC	Area Under Curve
CSV	Comma Separated Values
DL	Deep Learning
IoT	Internet of Things
ML	Machine Learning
MLP	Multilayer Perceptron
PDF	Portable Document Format
RF	Random Forest
ROC	Receiver Operating Characteristic
TDMA	Time Division Multiple Access
WSN-DS	Wireless Sensor Network Dataset
WSNs	Wireless Sensor Networks
XGBOOST	Extreme Gradient Boosting

CHAPTER-1

INTRODUCTION

1.1 Cybersecurity in Wireless Sensor Networks

Wireless Sensor Networks (WSNs) are widely used in modern technological applications such as environmental monitoring, industrial automation, healthcare systems, and smart infrastructure. These networks consist of multiple sensor nodes that collect and transmit data wirelessly to a central base station. Due to their distributed architecture and wireless communication nature, WSNs play a critical role in enabling real-time data acquisition and intelligent decision-making.

However, the open communication medium and limited hardware resources of WSNs make them highly vulnerable to cyberattacks. Unlike traditional networks, sensor nodes in WSNs have constrained processing power, memory, and battery life, which limits the implementation of complex security mechanisms. As a result, attackers can easily exploit these vulnerabilities to disrupt network operations, manipulate data, or degrade system performance.

Cybersecurity in WSNs focuses on protecting network communication, ensuring data integrity, and preventing unauthorized access or malicious activities. In such networks, maintaining secure communication is essential because any compromise can lead to incorrect data interpretation, system malfunction, or even complete network failure.

Various types of cyberattacks target WSNs due to their weak security infrastructure. Among the most common attacks are Blackhole attacks, Grayhole attacks, Flooding attacks, and Time Division Multiple Access (TDMA)-based attacks. These attacks exploit routing protocols, communication patterns, and node behavior to disrupt the network.

In a Blackhole attack, a malicious node falsely advertises itself as having the best route to the base station and then drops all incoming data packets. In a Grayhole attack, the malicious node selectively drops packets, making detection more difficult. Flooding attacks generate excessive network traffic, consuming bandwidth and energy resources. TDMA-based attacks interfere with scheduled communication slots, causing collisions and network instability.

These threats highlight the importance of developing robust cybersecurity solutions specifically tailored for Wireless Sensor Networks.

1.2 Cyberattack Detection using Machine Learning

Traditional security techniques such as encryption and authentication are primarily used to protect data confidentiality and prevent unauthorized access. However, these methods are not sufficient to detect or classify cyberattacks occurring within the network.

To address this limitation, machine learning and deep learning techniques have been widely adopted in cybersecurity. These techniques analyze network traffic patterns and identify anomalies that indicate malicious activity. By learning from historical data, machine learning models can distinguish between normal and abnormal network behavior.

Several machine learning models such as Random Forest and XGBoost have been successfully used for cyberattack detection. These models provide high accuracy in classification tasks and can handle large datasets effectively. Deep learning models such as Multilayer Perceptron (MLP) further enhance detection capabilities by capturing complex nonlinear relationships in network data. In addition to supervised learning methods, unsupervised learning techniques such as Autoencoders are used for anomaly detection. These models learn the normal behavior of the network and identify deviations that may indicate unknown or emerging attacks.

However, relying on a single model for cyberattack detection has several limitations. Different models perform differently depending on the type of attack and network conditions. Some models may perform well for certain attack types but fail for others. Additionally, single-model approaches may lack stability and robustness in dynamic environments.

To overcome these challenges, the concept of ensemble learning is introduced. Ensemble learning combines multiple models to improve overall performance. By integrating different models, the system can leverage their individual strengths and reduce weaknesses.

1.3 Objectives

The objective of this system is to is to develop an efficient and intelligent cyberattack detection and classification system for Wireless Sensor Networks using a unified ensemble learning approach. The system is designed to integrate multiple machine learning and deep learning models, including Random Forest, XGBoost, Multilayer Perceptron (MLP), and Autoencoder, to enhance detection performance. It aims to accurately detect and classify various types of cyberattacks such as Blackhole, Grayhole, Flooding, and TDMA-based attacks while also identifying unknown or emerging threats through anomaly detection techniques. Additionally, the project focuses on improving detection accuracy, classification stability, and overall system reliability. The proposed system supports real-time monitoring and analysis of network traffic and provides forensic logging along with actionable recommendations to help prevent future cyberattacks.

1.4 Problem Statement

Wireless Sensor Networks (WSNs) are highly vulnerable to cyberattacks due to their limited resources and open wireless communication. Existing intrusion detection systems often rely on single models, which cannot accurately classify multiple attack types or detect unknown threats. They also struggle to adapt to dynamic network conditions and lack real-time monitoring capabilities. Therefore, there is a need for an efficient system that can accurately detect and classify various cyberattacks, identify new threats, and provide reliable performance, which is achieved using the proposed unified ensemble learning approach.

CHAPTER-2

LITERATURE SURVEY

[1]. This paper presents a machine learning-based approach for detecting cyberattacks in Wireless Sensor Networks. The study focuses on identifying abnormal network behavior using classification algorithms and highlights the effectiveness of machine learning techniques in improving detection accuracy. However, the system primarily focuses on detection and does not provide detailed classification of multiple attack types.

[2]. This study proposes a predictive cybersecurity framework using machine learning and deep learning techniques for intrusion detection in WSN environments. The system improves detection performance by analyzing network traffic patterns, but it lacks adaptability to dynamic network conditions and does not effectively handle unknown or emerging attacks.

[3]. This paper explores intrusion detection in IoT and WSN environments using machine learning algorithms. The model analyzes network traffic to identify malicious activities and improve system security. Although the approach enhances detection capability, it does not provide consistent performance across different attack types.

[4]. This research provides a detailed analysis of the integration of machine learning techniques in Wireless Sensor Network security. The study highlights how different algorithms can improve attack detection and classification. However, it also emphasizes the challenges related to computational complexity and scalability in real-time environments.

[5]. This paper investigates machine learning-based detection of network security incidents in Wireless Sensor Networks. The system demonstrates good performance in identifying attacks but shows limitations in adapting to varying network conditions and evolving attack patterns.

[6]. This study proposes a hybrid feature reduction technique combined with machine learning for cyberattack detection. The approach improves detection efficiency by reducing redundant data features. However, it does not incorporate

advanced ensemble techniques for improving classification stability.

[7]. This research focuses on enhancing WSN security using machine learning-based DoS attack detection. The system effectively detects denial-of-service attacks but is limited to specific attack types and lacks a generalized multi-attack detection framework.

[8]. This paper presents a machine learning and deep learning-based intrusion detection system for cybersecurity applications. The approach improves detection accuracy using advanced models but requires high computational resources, making it less suitable for resource-constrained WSN environments.

[9]. This study introduces a machine learning-based intrusion detection system for Wireless Sensor Networks that analyzes network behavior to identify malicious activity. While the system improves detection accuracy, it does not effectively classify multiple attack categories.

[10]. This research focuses on detecting network security incidents using machine learning models and highlights the importance of analyzing traffic patterns for identifying attacks. However, the system lacks integration of multiple models to enhance performance.

CHAPTER-3

DESIGN

3.1 Existing System

In Wireless Sensor Networks, existing cyberattack detection systems mainly rely on traditional machine learning or deep learning models to identify malicious activities. These systems typically analyze network traffic data and classify it as normal or attack based on learned patterns. Common approaches include the use of algorithms such as Random Forest, XGBoost, Support Vector Machines, and neural networks.

Most of the existing systems focus only on detecting whether an attack is present in the network, without accurately classifying the type of attack. This limitation reduces the effectiveness of the system in real-world applications where identifying the exact attack type is crucial for implementing appropriate countermeasures.

Additionally, many systems rely on a single machine learning model, which leads to inconsistent performance across different types of attacks. Some models may perform well for specific attack types but fail to generalize across all scenarios. Furthermore, deep learning-based systems require high computational resources and are not suitable for resource-constrained environments like Wireless Sensor Networks.

Another major limitation of existing systems is their inability to detect unknown or emerging cyberattacks. Most systems are trained only on known attack patterns and fail to identify new threats that do not match the training data. Moreover, many systems lack real-time monitoring capabilities and do not provide detailed analysis or recommendations for preventing future attacks.

3.1.1 Disadvantages of Existing System

- **Limited Attack Classification:** Most existing systems are designed only to detect whether an attack is present or not, but they do not accurately classify different types of attacks such as Blackhole, Grayhole, Flooding, and TDMA. This limitation reduces the effectiveness of the system in real-world scenarios where identifying the type of attack is essential for applying proper countermeasures.

- **Dependence on Single Model:** Many existing intrusion detection systems rely on a single machine learning or deep learning model. As a result, the system performance becomes inconsistent because different models perform differently for various attack types. This lack of model diversity reduces the overall reliability and robustness of the system.

- **Inability to Detect Unknown Attacks:** Most systems are trained using predefined datasets containing known attack patterns. These systems fail to detect new or emerging cyber threats that do not match the training data, making them ineffective against evolving attack strategies.

- **Lack of Adaptability:** Existing systems are not capable of adapting to dynamic network conditions. Changes in network traffic patterns, node behavior, or attack strategies can significantly affect system performance, leading to inaccurate results.

- **No Real-Time Monitoring:** Many existing approaches do not support real-time monitoring and analysis of network traffic. This delay in detection reduces the system's ability to respond quickly to cyberattacks.

3.2 Proposed System

To overcome the limitations of existing systems, this project proposes a **Unified Stacking Ensemble Learning Framework** for cyberattack detection and classification in Wireless Sensor Networks.

The proposed system integrates multiple machine learning and deep learning models, including Random Forest, XGBoost, Multilayer Perceptron (MLP), and Autoencoder, into a single unified framework. This combination allows the system to leverage the strengths of each model and improve overall detection accuracy and reliability.

The system uses supervised learning techniques to classify known cyberattacks and unsupervised learning techniques to detect unknown or emerging threats. Random Forest and XGBoost models are used for robust classification of known attacks, while the MLP model captures complex nonlinear relationships in network data. The Autoencoder is used for anomaly detection by identifying deviations from normal network behavior.

The proposed system is trained using the WSN-DS dataset, which contains various types of network traffic data, including normal and malicious activities. Additionally, the system is evaluated using network traffic generated from the COOJA simulation environment to ensure realistic testing conditions.

The use of stacking ensemble learning enables the system to combine the outputs of individual models and produce a final prediction with improved accuracy and stability. This approach enhances the system's ability to detect complex attack patterns and adapt to dynamic network environments.

3.2.1 System Architecture

The system architecture of the proposed unified ensemble learning framework is designed to efficiently detect and classify cyberattacks in WSNs. The architecture provides a structured approach for processing network data, applying machine learning models, and generating accurate detection results. It ensures smooth interaction between different components of the system and supports real-time monitoring and analysis of cyber threats.

The proposed system architecture mainly consists of three major modules: the User Interface module, the Cyberattack Detection Engine, and the Data Storage module. These modules work together to perform data processing, attack detection, classification, and result storage in an organized manner.

Initially, the system receives input data from two main sources: the WSN-DS dataset and the network traffic generated using the COOJA simulation environment. These datasets include both normal and malicious network behavior, which are essential for training and evaluating the detection models.

Once the data is collected, it is passed to the preprocessing stage, where several operations such as data cleaning, normalization, and feature selection are performed. Data cleaning removes unnecessary or inconsistent values, normalization scales the data for better model performance, and feature selection identifies the most relevant attributes for analysis. These preprocessing steps improve the quality of data and enhance the accuracy of the models.

After preprocessing, the data is analyzed using multiple machine learning and deep learning models, including Random Forest, XGBoost, and Multilayer Perceptron (MLP). Each model processes the data independently and generates predictions based on learned patterns. These outputs are combined using a stacking ensemble learning approach to produce a final classification result with improved accuracy and stability.

In addition, an Autoencoder model is used for anomaly detection to identify unknown or emerging cyberattacks by detecting deviations from normal network behavior. The final output includes attack detection, classification of attack types, and analysis results, which are stored for future reference and security monitoring.

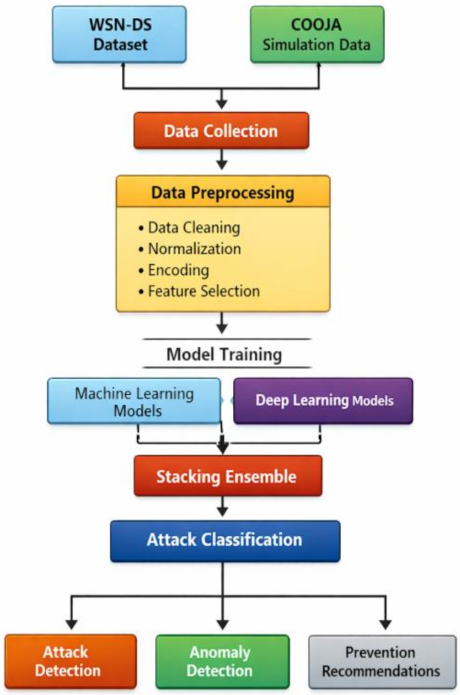


Fig 3.1 : System Architecture

3.3 System Modules

The proposed system is divided into several modules to ensure efficient processing, accurate detection, and proper management of cyberattack data in Wireless Sensor Networks. Each module performs a specific function and collectively contributes to the overall working of the system.

3.3.1 User Interface Module

The User Interface module provides interaction between the user and the system. It allows users to upload datasets, initiate the cyberattack detection process, and view the results generated by the system. This module also supports user authentication to ensure secure access. Additionally, users can access information such as detected attack types, anomaly alerts, and prevention recommendations through this interface.

3.3.2 Cyberattack Detection Engine

The Cyberattack Detection Engine is the core component of the system, responsible for processing data and detecting cyberattacks. It performs preprocessing steps such as data cleaning, normalization, encoding, and feature selection to improve data quality. After preprocessing, it applies machine learning models like Random Forest and XGBoost, along with deep learning models such as Multilayer Perceptron, to analyze network traffic.

The outputs from these models are combined using stacking ensemble learning to produce accurate classification results. Additionally, an Autoencoder is used for anomaly detection to identify unknown or emerging cyberattacks. This module ensures reliable and efficient detection of both known and unknown threats.

3.3.3 Data Storage Module

The Data Storage module is responsible for storing all relevant data, including input datasets, processed data, detection results, and forensic logs. It maintains records of detected attacks, classification results, and system performance metrics. These stored logs help in analyzing past incidents and support longterm monitoring of network security. The module also enables easy retrieval of data for further analysis and reporting.

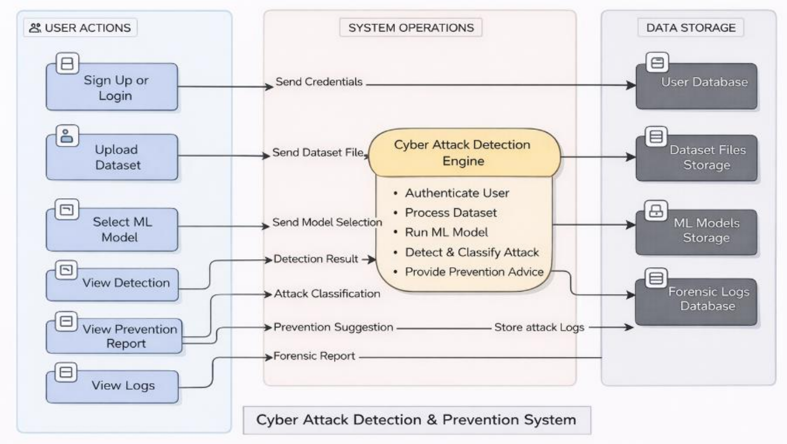


Fig 3.2: System Modules of the proposed system

3.4 Need for Unified Ensemble Model

In traditional cyberattack detection systems, a single machine learning model is used, which limits the ability to detect and classify different types of attacks accurately. Each model performs well for specific patterns but fails to provide consistent results in dynamic network conditions. Wireless Sensor Networks require a more reliable approach due to varying traffic and evolving attack patterns.

To overcome these limitations, the proposed system uses a unified ensemble learning approach that combines multiple models such as Random Forest, XGBoost, Multilayer Perceptron, and Autoencoder. This integration improves detection accuracy, enhances stability, and enables the system to identify both known and unknown cyberattacks effectively.

3.5 Dataset Processing Workflow

3.5.1 Training Dataset

The dataset used to train the proposed model is the WSN-DS dataset, which consists of 374,661 samples of network communication including both normal traffic and cyberattacks. The dataset contains various types of attacks such as Blackhole, Grayhole, Flooding, and TDMA attacks. Important network parameters used for training include packet transmission rate, routing path information, and energy consumption of nodes.

The dataset captures key network communication properties such as routing behavior, packet transmission patterns, cluster head associations, communication distance, and node power usage. These parameters play a significant role in identifying abnormal node behavior and detecting cyberattacks.

The system should perform secure key exchange using ECC between the sender and receiver.

To evaluate the model effectively, the dataset is divided into training and testing subsets. The unified ensemble model integrates Random Forest, XGBoost, Multilayer Perceptron (MLP), and Autoencoder within a single framework, which improves detection accuracy, enhances classification stability, and enables anomaly detection in Wireless Sensor Networks.

3.5.2 Simulation Environment

To validate the performance of the proposed unified ensemble model under realistic conditions, additional network traffic is generated using the COOJA network simulator, which is part of the Contiki-NG operating system. COOJA is a realistic simulation environment that mimics sensor node hardware behavior, wireless communication protocols, and radio transmission characteristics.

The simulator models various aspects of network behavior, including packet transmission, node energy consumption, and network topology. It supports both normal network communication and malicious node behavior, making it suitable for testing cyberattack detection systems in real-world WSNs scenarios.

In the simulation setup, sensor nodes are randomly deployed within a defined area, and a base station is placed centrally to collect data. The nodes communicate using multi-hop routing, where data is transmitted through intermediate nodes to ensure reliable communication. The network follows IEEE 802.15.4 communication standards, commonly used in WSN and IoT applications.

Initially, the simulation runs under normal conditions to establish baseline network behavior, including packet transmission rate, routing patterns, communication delay, and energy consumption. After establishing normal behavior, cyberattack scenarios are introduced by modifying the behavior of selected nodes. This helps in analyzing how attacks affect network performance and communication stability.

Settings	Values Used In Simulation
Node Type	Cooja sensor mote
Number Of Nodes	200 nodes
Node Distribution	Random deployment in 200m x 200m area
Simulation Time	900 seconds
Radio Medium	UDGM (Unit Disk Graph Medium)
Transmission Range	50 meters
MAC Protocol	IEEE 802.15.4
Communication Protocol	UDP
Packet Transmission Interval	5 seconds
Energy Consumption Monitoring	Enabled (per-node energy tracking)
Network Traffic Logs	Mote output logs converted to CSV format

Table 3.1 : COOJA Simulation Settings

The simulation generates network communication logs at the mote output level, which include details of the network. These logs are processed using scripts and converted into CSV format for further analysis. The processed simulation data is then used to test and validate the performance of the proposed system.

3.6 Methodology

The proposed methodology is based on a unified ensemble learning approach for detecting and classifying cyberattacks in Wireless Sensor Networks. Instead of relying on a single model, the system integrates multiple machine learning and deep learning models to analyze different patterns of network traffic and improve overall detection performance.

The methodology uses Random Forest, XGBoost, and Multilayer Perceptron (MLP) as supervised learning models for classifying known cyberattacks. Random Forest is selected due to its ability to handle noisy and high-dimensional data, which is common in Wireless Sensor Network traffic. It provides stable and reliable classification results by constructing multiple decision trees and combining their outputs.

Multilayer Perceptron (MLP), a deep learning model, is used to learn nonlinear relationships present in network traffic data. Wireless Sensor Network communication often involves complex interactions between parameters such as packet transmission rate, routing behavior, and energy consumption. MLP captures these nonlinear patterns and improves the system's ability to classify different attack types.

Autoencoder is used for anomaly detection in the proposed system. It is trained using normal network traffic and learns the standard behavior of the network. When abnormal or unseen data is encountered, the model detects deviations based on reconstruction error, allowing the system to identify unknown or emerging cyberattacks that are not present in the training dataset.

The outputs of the supervised models (Random Forest, XGBoost, and MLP) are combined using stacking ensemble learning. In this approach, the predictions of individual models are used as input to a final classifier, which produces the final classification result. This method improves detection accuracy, reduces false positives and false negatives, and enhances classification stability.

3.7 Algorithm: Unified Ensemble Framework for Cyberattack Detection

The proposed algorithm describes the step-by-step process of detecting and classifying cyberattacks in Wireless Sensor Networks using a unified ensemble learning approach. It integrates multiple machine learning and deep learning models along with anamoly detection to improve accuracy and reliability.

1. *Dataset Preparation:*

   Split processed dataset D_{proc} into training and testing sets:

   $$\{\boldsymbol{D_{train}}, \boldsymbol{D_{test}}\} = \boldsymbol{Split}(\boldsymbol{D_{proc}}) \tag{1}$$

2. *Model Initialization*:

   Base classifiers: Random Forest, XG Boost (XGB), Multilayer Perceptron (MLP) and Autoencoder (AE) is used for anomaly detection.

3. *Model Training:*

   Train RF, XGB, and MLP using D_{train}

   Train AE using only normal traffic samples $D_{normaltrain}$.

4. *Meta-Feature Generation:*

   Generate base model outputs

   $$\boldsymbol{MF_{train}} = \{\, \mathbf{RF}(\mathbf{x}), \mathbf{XGB}(\mathbf{x}), \mathbf{MLP}(\mathbf{x})\}, \forall \mathbf{x} \in \boldsymbol{D_{train}}. \tag{2}$$

5. *Meta-Learner Training:*

   Train stacking classifier using MF_{train} and labels Y_{train}

6. *Prediction:*

   Generate meta-features MF(x)MF(x)MF(x) and obtain final classification

   $$\boldsymbol{C}(\boldsymbol{x}) = \boldsymbol{Meta}(\boldsymbol{MF}(\boldsymbol{x})) \tag{3}$$

7. *Anomaly Detection:*

   The Autoencoder computes the reconstruction error:

   $$\boldsymbol{RE}(\boldsymbol{x}) = ||\ \boldsymbol{x} - \boldsymbol{AE}(\boldsymbol{x})\ ||^{\mathbf{2}} \tag{4}$$

8. *Threshold Calculation:*

   The anomaly threshold is calculated using training reconstruction error statistics

   $$\boldsymbol{\tau} = \boldsymbol{\mu}(\boldsymbol{REtrain}) + \boldsymbol{k\sigma}(\boldsymbol{REtrain}) \tag{5}$$

9. *Unknown Attack Detection:*

   If $RE(x) > \tau$, the input is classified as an unknown attack; otherwise, it is treated as normal or classified using the ensemble model.

10. *Final Decision:*

    If an anomaly is detected, the output is **Unknown Attack**; otherwise, the ensemble classifier output $C(x)$ is returned.

3.8 Attack Detection Mechanism

The attack detection mechanism in the proposed system is designed to identify, classify, and analyze cyberattacks in Wireless Sensor Networks using both supervised and unsupervised learning techniques. The system processes network traffic data and detects abnormal patterns based on trained models.

Initially, the data is analyzed using machine learning models such as Random Forest, XGBoost, and Multilayer Perceptron, which classify the traffic into normal or specific attack categories such as Blackhole, Grayhole, Flooding, and TDMA attacks. These models are combined using stacking ensemble learning to improve detection accuracy and classification stability.

In addition to classification, the system uses an Autoencoder model for anomaly detection. The Autoencoder learns normal network behavior and identifies deviations based on reconstruction error. If the error exceeds a predefined threshold, the system classifies the input as an unknown or emerging attack.

The system also identifies malicious nodes responsible for attacks by analyzing abnormal communication patterns and network behavior. Along with detection, the system generates prevention recommendations specific to each type of attack, which helps users take appropriate security measures.

All detected attacks and malicious node information are stored in forensic logs for future analysis. The system provides filtering options that allow users to view attack statistics based on time intervals such as today, yesterday, this month, and this year. Additionally, users can export these reports in PDF or CSV format for documentation and further analysis.

CHAPTER-4

IMPLEMENTATION

4.1 System Configuration

The implementation of the proposed cyberattack detection system requires both hardware and software components to ensure efficient performance.

4.1.1 Hardware Configuration

The hardware configuration used for implementing the system is listed below:

- **Processor:** Intel Core i5 or equivalent
- **RAM:** Minimum 8 GB
- **Storage:** 256 GB Hard Disk / SSD
- **System Type:** 64-bit operating system
- **Network:** Standard internet connectivity

4.1.2 Software Configuration

The software tools used for developing and implementing the system include the following:

- **Operating System:** Windows / Linux
- **Programming Language:** Python, C
- **Development Tool:** Jupyter Notebook / VS Code
- **Database:** MySQL / SQLite
- **Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow / Keras
- **Simulation Tool:** COOJA Simulator

- **Dataset:** WSN-DS Dataset

4.2 Tools and Technologies Used

The system is implemented using Python due to its extensive support for machine learning and data analysis. Libraries such as NumPy and Pandas are used for data preprocessing, while Scikit-learn is used for implementing machine learning models like Random Forest and XGBoost. TensorFlow or Keras is used for implementing deep learning models such as Multilayer Perceptron and Autoencoder.

The COOJA simulator is used to generate realistic Wireless Sensor Network traffic, including both normal and malicious scenarios. Jupyter Notebook or Visual Studio Code is used as the development environment for implementing and testing the system.

4.3 Dataset Implementation

The implementation uses two datasets: the WSN-DS dataset for training and the data generated from the COOJA simulator for testing and validation. The dataset is loaded into the system using Python libraries such as Pandas.

Preprocessing steps such as data cleaning, normalization, encoding, and feature selection are applied to improve data quality. The dataset is then split into training and testing sets to evaluate model performance. The training dataset is used to train the models, while the testing dataset is used to validate the system under realistic conditions.

4.4 Model Implementation

The proposed system implements multiple machine learning and deep learning models for cyberattack detection. Random Forest is implemented using the Scikit-learn library and is used for handling noisy data and providing stable classification results. XGBoost is implemented using the XGBoost library and is used to improve prediction accuracy by capturing complex patterns in network traffic.

The Multilayer Perceptron (MLP) model is implemented using TensorFlow or Keras to learn nonlinear relationships in the data. The Autoencoder model is also implemented using deep learning frameworks and is trained only on normal network traffic to detect anomalies based on reconstruction error.

Each model is trained individually using the training dataset, and their outputs are later combined using stacking ensemble learning to improve overall detection performance.

4.5 System Workflow Implementation

The system workflow is implemented using Python to handle data processing, model execution, attack detection, and result management. Initially, the dataset is loaded using the Pandas library and preprocessed using functions for data cleaning, normalization, encoding, and feature selection.

The trained machine learning models such as Random Forest, XGBoost, and Multilayer Perceptron are loaded and used to generate predictions for input network traffic data. The outputs from these models are combined using a stacking ensemble method implemented through a meta-classifier to produce the final classification result. The Autoencoder model is used to compute reconstruction error, and a threshold value is applied to identify anomalies.

For malicious node detection, the system analyzes network parameters such as node ID, packet transmission behavior, routing patterns, and abnormal communication frequency. Nodes showing abnormal behavior or participating in attack patterns are flagged as malicious and stored in the system.

Prevention recommendations are implemented using predefined rule-based mappings, where each detected attack type is associated with specific prevention strategies. Based on the classification output, the system retrieves and displays the corresponding prevention measures.

All detected attacks and malicious node details are stored in a structured database (such as SQLite or JSON-based storage). Each record includes attributes such as attack type, node ID, timestamp, and severity level.

To support filtering functionality, queries are implemented to retrieve attack logs based on time intervals such as current day, previous day, monthly, and yearly data. This is achieved using timestamp-based filtering logic.

For report generation, the system uses Python libraries such as Pandas and ReportLab (or CSV writer) to export the filtered data into PDF and CSV formats. The exported reports include attack details, malicious nodes, and statistical summaries.

CHAPTER-5

TESTING

5.1 Testing Approach

In this project, testing is carried out to verify the correctness and reliability of the proposed cyberattack detection system. Two types of testing are performed, namely functional testing and performance testing. Functional testing ensures that all modules such as data preprocessing, model execution, anomaly detection, malicious node detection, and report generation are working correctly. Performance testing evaluates the efficiency of the system in detecting cyberattacks and generating outputs within acceptable time.

5.2 Features to be Tested

After implementing the complete system, the following features are tested to ensure proper functionality:

- Verify that the system correctly loads and preprocesses the dataset.
- Ensure that machine learning models correctly classify network traffic into normal and attack categories.
- Check whether ensemble learning improves classification performance.
- Verify that the Autoencoder detects unknown attacks effectively.
- Ensure that malicious nodes are correctly identified.
- Check whether attack logs and reports are generated correctly.

5.3 Testing tools and environment

- **Programming Environment:** Visual Studio Code / Jupyter Notebook used to develop and test the cyberattack detection system.
- **Programming Language:** Python – used to implement data preprocessing, machine learning models, and anomaly detection.

- **Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow/Keras – used for data handling, model training, and evaluation.
- **Database:** MySQL / SQLite - it stores user credentials, store attack logs, malicious node information, and generated reports.
- **Dataset:** WSN-DS Dataset – used for training and evaluating the detection models.
- **Simulation Tool:** COOJA Simulator – used to generate realistic Wireless Sensor Network traffic for testing.

5.4 Test cases

5.4.1 Input

This project requires the following inputs:

1. **User authentication credentials:** Username and password are required for user authentication to securely access the system.
2. **Dataset input**: The user uploads the dataset to analyze network traffic and detect cyberattacks.

5.4.2 Expected Output

The following outcomes are expected from the proposed system:

- Successful authentication of users using valid credentials.
- Accurate classification of network traffic into normal and different attack types.
- Detection of cyberattacks such as Blackhole, Grayhole, Flooding, and TDMA attacks.
- Identification of malicious nodes involved in the attacks.
- Detection of unknown or new attacks using anomaly detection.
- Generation of prevention recommendations based on detected attacks.

- Storage of attack details and malicious node information in forensic logs.
- Correct filtering of attack data based on time intervals such as today, yesterday, monthly, and yearly.
- Successful export of reports in PDF and CSV formats.

5.4.3 Testing Procedure

The following procedure is used to test the proposed system:

- Successful login of the user with access to system modules such as Upload, Algorithm, Dashboard, Forensics, and View Data.
- In the Upload section, the system accepts the dataset and performs cyberattack detection, providing prediction results.
- The uploaded dataset is displayed in the View Data section for user verification and analysis.
- The Dashboard displays dataset analytics, attack statistics, and identified malicious nodes
- The Algorithm section shows performance metrics of the unified ensemble model, including accuracy, classification results, and confusion matrix.
- The Forensics section stores attack logs, including detected attacks and malicious node information for future analysis.

CHAPTER-6

PERFORMANCE METRICS

The performance of the proposed **Unified Ensemble Learning based Cyberattack Detection System** is evaluated using standard performance metrics. These metrics help in analyzing how effectively the system detects and classifies cyberattacks in WSNs.

The evaluation is carried out using the testing dataset, and the results are measured based on metrics such as Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC Curve. These metrics provide a clear understanding of the model's performance in distinguishing between normal network traffic and different types of cyberattacks.

6.1 Accuracy

Accuracy is used to measure the overall correctness of the system. It indicates the percentage of correctly classified instances out of the total number of instances.

Accuracy is calculated using the following formula:

$$\frac{(\mathbf{TP} + \mathbf{TN})}{(\mathbf{TP} + \mathbf{TN} + \mathbf{FP} + \mathbf{FN})} \qquad \textbf{(1)}$$

Where TP represents correctly detected attack instances, TN represents correctly detected normal instances, FP represents normal traffic that is wrongly classified as an attack, and FN represents attack instances that are wrongly classified as normal.

The proposed unified ensemble model achieves an accuracy of **99.88%**, which shows that the system performs highly effectively in detecting cyberattacks.

6.2 Precision

Precision measures how accurately the system predicts attack instances among all predicted attack cases.

Precision is calculated using the following formula:

$$\frac{\boldsymbol{TP}}{(\mathbf{TP} + \mathbf{FP})} \tag{2}$$

A high precision value indicates that the system produces fewer false alarms.

The proposed model achieves a precision value of approximately **0.99**, indicating high reliability in prediction.

6.3 Recall

Recall, also known as detection rate, measures the ability of the system to correctly identify actual attack instances.

Recall is calculated using the following formula:

$$\frac{\boldsymbol{TP}}{(\mathbf{TP} + \mathbf{FN})} \tag{3}$$

The proposed system achieves a recall value of **0.99**, which shows that it can effectively detect most of the cyberattacks present in the network.

6.4 F1-Score

F1-Score is used to measure the balance between precision and recall. It is the harmonic mean of precision and recall.

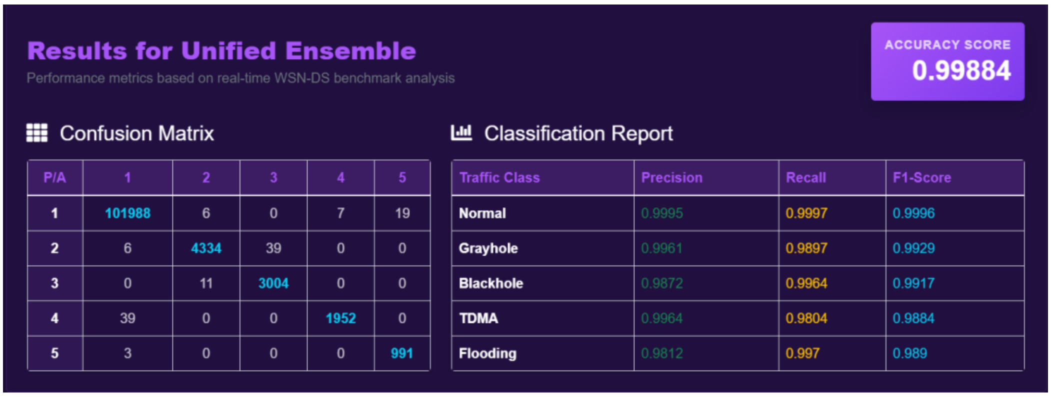
F1-Score is calculated using the following formula:

$$\mathbf{2} \times \frac{(\mathbf{Precision} \times \mathbf{Recall})}{(\mathbf{Precision} + \mathbf{Recall})} \tag{4}$$

The proposed model achieves an F1-Score of approximately **0.99**, which indicates a balanced and consistent performance.

6.5 Confusion Matrix

The confusion matrix is used to evaluate the classification performance by comparing actual and predicted values. It shows correct classifications and misclassification cases. Most values lie along the diagonal, indicating that the model correctly classifies the majority of instances, which proves its effectiveness.


Results for Unified Ensemble

Performance metrics based on real-time WSN-DS benchmark analysis

ACCURACY SCORE 0.99884

Confusion Matrix

P/A	1	2	3	4	5
1	101988	6	0	7	19
2	6	4334	39	0	0
3	0	11	3004	0	0
4	39	0	0	1952	0
5	3	0	0	0	991

Classification Report

Traffic Class	Precision	Recall	F1-Score
Normal	0.9995	0.9997	0.9996
Grayhole	0.9961	0.9897	0.9929
Blackhole	0.9872	0.9964	0.9917
TDMA	0.9964	0.9804	0.9884
Flooding	0.9812	0.997	0.989

Fig. 6.1 : Confusion Matrix and Classification Report of Proposed System

6.6 ROC Curve

The ROC curve is used to evaluate the model's ability to distinguish between normal and attack classes. The AUC represents the overall performance of the model, where an AUC value close to 1 indicates excellent performance and an AUC value close to 0.5 indicates poor performance.

The proposed unified ensemble model achieves an AUC value of **0.98**, which is higher than individual models, showing better classification performance.

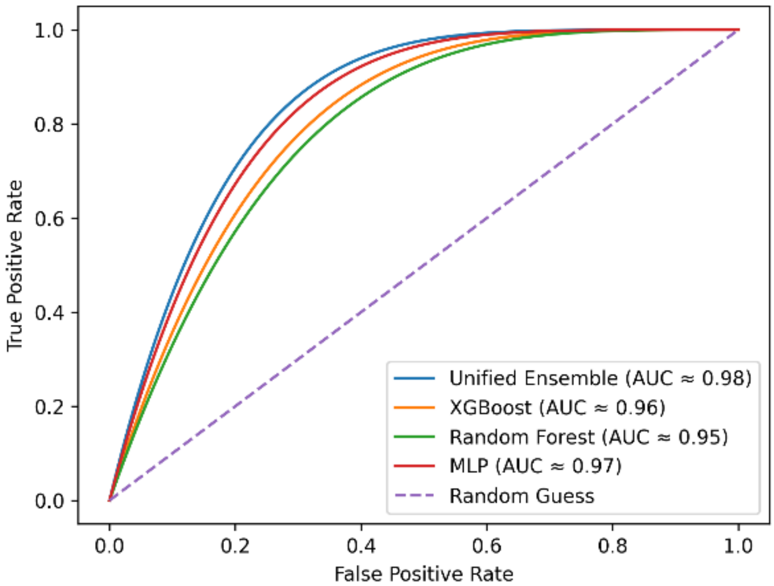


Fig. 6.2 : ROC Curve of the Proposed Model

6.7 Comparative Analysis

The performance of the proposed unified ensemble model is compared with individual machine learning models.

Model	**Accuracy (%)**	**Precision**	**Recall**	**F1- Score**
Unified Ensemble	99.88	0.99	0.99	0.99
XG Boost	99.73	0.95	0.99	0.97
Random Forest	99.71	0.93	0.99	0.96
Multilayer Perceptron	90.76	0.18	0.20	0.19
Autoencoder	99.12	0.98	0.99	0.98

Table 6.1 : Performance Comparison of Proposed Unified Ensemble Model with Individual Models

From the above table, it is observed that the proposed unified ensemble model achieves the highest accuracy, precision, recall, and F1-score compared to individual machine learning models. This demonstrates that combining multiple models using ensemble learning improves overall detection performance, classification stability, and reliability in identifying cyberattacks in Wireless Sensor Networks.

CHAPTER-7

RESULTS

The proposed **Unified Ensemble Learning** based cyberattack detection system was successfully implemented and tested using the WSN-DS dataset and network traffic generated from the COOJA simulation environment. The system is designed to detect and classify different types of cyberattacks in Wireless Sensor Networks and provide real-time analysis.

The system allows users to upload network datasets and perform cyberattack detection through the user interface. After processing the data, the system classifies the network traffic into normal and different attack types such as Blackhole, Grayhole, Flooding, and TDMA attacks. The results are displayed to the user along with prediction outputs and analysis.

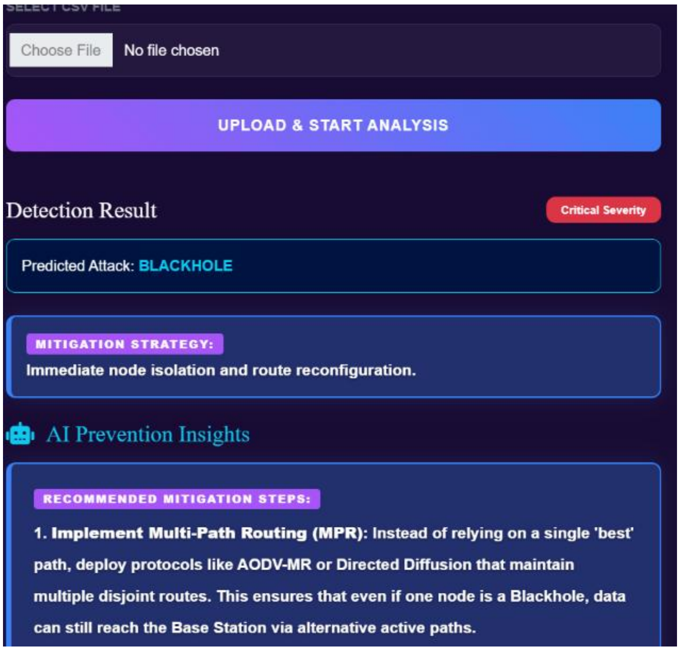


Fig 7.1: Cyberattack Detection Output Showing Predicted Attack and Mitigation Strategies

The system provides a manual input interface that allows users to enter network parameters such as node ID, communication details, energy levels, and routing information. The entered data is processed by the proposed unified ensemble model to detect potential cyberattacks. This functionality enables flexible testing and supports real-time analysis of different network scenarios.

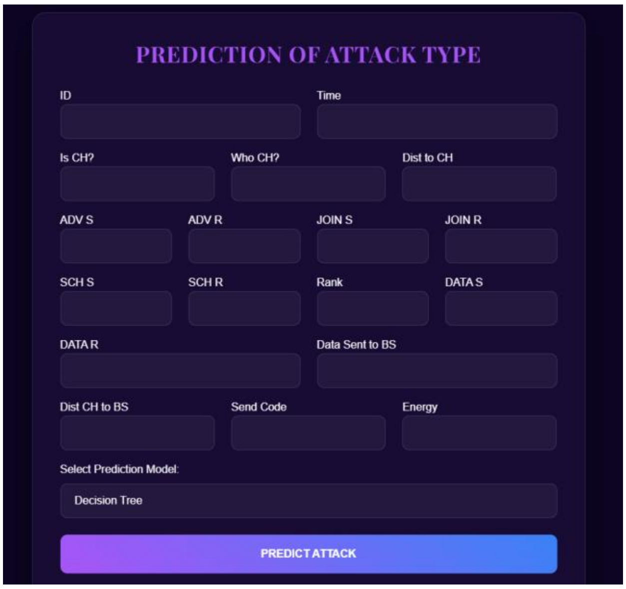


Fig 7.2: Manual Input Interface for Cyberattack Prediction

The uploaded dataset is displayed in the view data section, allowing users to examine the network traffic data in detail. It includes various parameters such as node ID, communication attributes, transmission values, and routing information. This helps in verifying the input data and understanding the features used for cyberattack detection.

Network Traffic Dataset

ID	Time	Is_CH	Who CH	Dist_To_CH	ADV_S	ADV_R	JOIN_S	JOIN_R	SCH_S	SCH_R	Rank	DATA_S	DATA_R	Data_Sent_To_BS	dist_CH_T
39	00:00.026	0.0	37.0	94.97	62.0	93.0	15.0	37.0	39.0	21.0	31.89	155.0	175.0	180.0	88.77
39	00:00.026	0.0	29.0	53.31	164.0	58.0	85.0	22.0	148.0	135.0	5.30	78.0	9.0	63.0	25.33
39	00:00.026	0.0	42.0	2.00	170.0	156.0	170.0	92.0	26.0	42.0	50.09	79.0	127.0	134.0	20.00
39	00:00.026	1.0	24.0	64.41	100.0	60.0	49.0	144.0	139.0	196.0	3.19	1.0	143.0	191.0	59.59
39	00:00.026	0.0	18.0	89.81	28.0	157.0	145.0	122.0	155.0	17.0	9.02	11.0	36.0	99.0	65.94
39	00:00.026	0.0	13.0	3.68	97.0	71.0	141.0	113.0	160.0	183.0	98.44	180.0	32.0	140.0	35.21
39	00:00.026	1.0	46.0	39.85	156.0	166.0	153.0	144.0	64.0	129.0	11.25	191.0	196.0	155.0	77.92
39	00:00.026	1.0	32.0	20.30	157.0	70.0	67.0	153.0	126.0	87.0	40.05	142.0	61.0	160.0	11.94
39	00:00.026	1.0	5.0	52.06	179.0	55.0	71.0	71.0	124.0	132.0	97.86	6.0	119.0	95.0	97.96
34	00:00.027	0.0	36.0	93.33	44.0	160.0	37.0	19.0	37.0	32.0	48.69	180.0	193.0	63.0	47.22
34	00:00.027	1.0	18.0	58.11	98.0	89.0	110.0	196.0	154.0	18.0	27.92	91.0	77.0	69.0	53.94
34	00:00.027	1.0	34.0	9.02	34.0	184.0	30.0	77.0	156.0	77.0	86.91	125.0	160.0	79.0	52.32
34	00:00.027	1.0	7.0	10.20	56.0	57.0	64.0	88.0	47.0	16.0	46.79	63.0	47.0	49.0	66.66

Fig 7.3: View Data Section Displaying Uploaded Network Traffic Dataset

The dashboard provides a clear visualization of dataset analytics, attack statistics, and malicious node identification. It helps users to understand the distribution of attacks and analyze network behavior effectively. The system also identifies malicious nodes based on abnormal communication patterns and network parameters.

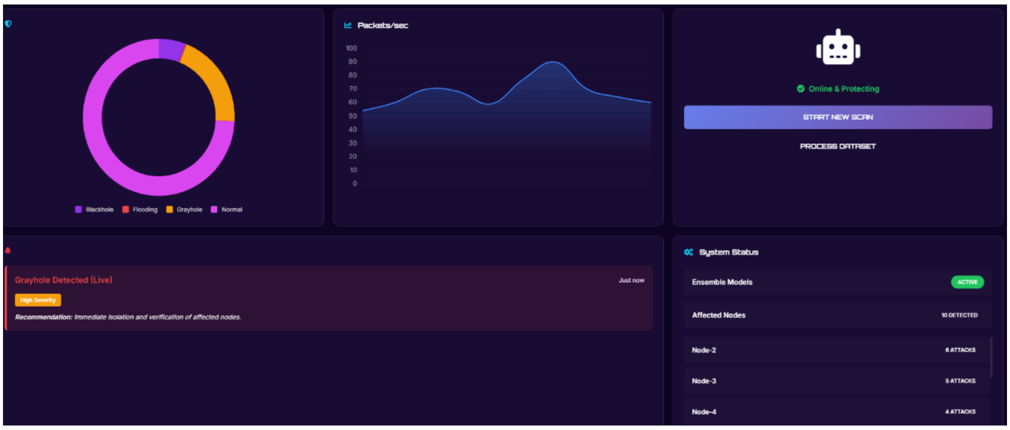


Fig 7.4: System Dashboard Displaying Attack Detection and Network Analytics

The system also includes a forensics module, where all detected attacks and malicious node details are stored as logs. These logs can be filtered based on time intervals such as daily, monthly, and yearly data. This helps in analyzing past attacks and improving network security.

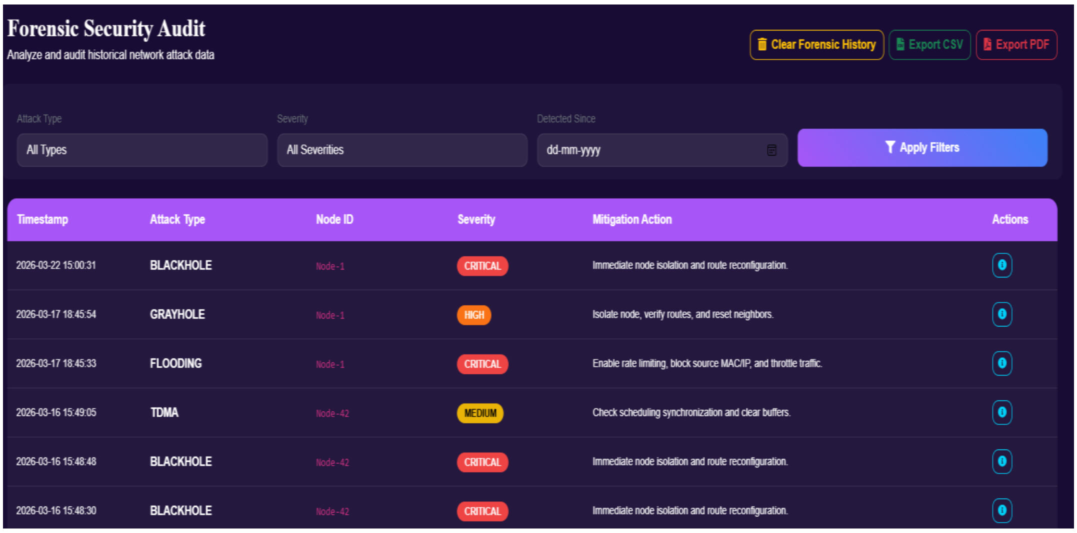


Fig 7.5: Forensic Module Showing Attack Logs and Mitigation Details

The system visualizes attack frequency trends and node risk distribution to provide better insights into network behavior. The attack frequency graph shows the variation of detected attacks over time, helping to identify patterns and peak attack periods. The node risk summary represents the distribution of risk levels such as critical, high, medium, and low across the network. This helps in understanding the severity of attacks and prioritizing security measures effectively.

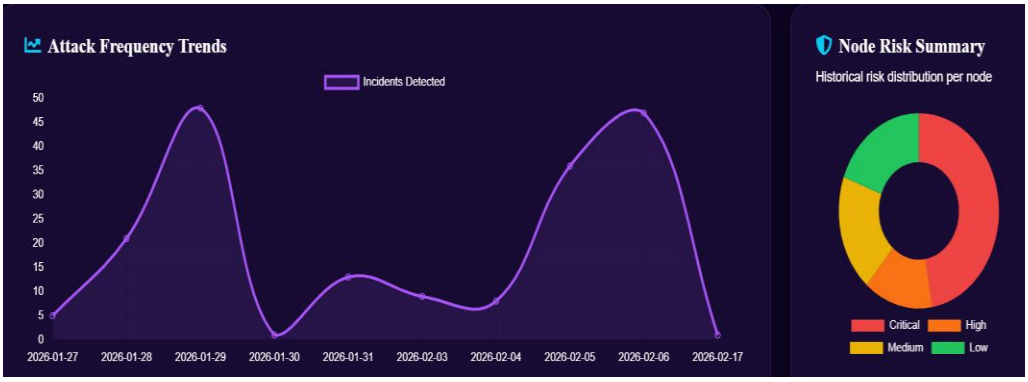


Fig 7.6: Attack Frequency Trends and Node Risk Distribution

The forensic attack logs can be exported as a PDF report for easy documentation. The report includes important details such as timestamp, attack type, node ID, severity, and mitigation actions. This helps in keeping proper records of attacks and makes it easier to analyze and review security incidents.

CyberDetect AI - Forensic Security Audit

Generated on: 2026-03-22 15:15:38

Timestamp	Attack Type	Node ID	Severity	Mitigation Action
2026-03-22 15:00:31	BLACKHOLE	Node-1	Critical	Immediate node isolation and route reconfigur...
2026-03-17 18:45:54	GRAYHOLE	Node-1	High	Isolate node, verify routes, and reset neighb...
2026-03-17 18:45:33	FLOODING	Node-1	Critical	Enable rate limiting, block source MAC/IP, an...
2026-03-16 15:49:05	TDMA	Node-42	Medium	Check scheduling synchronization and clear bu...
2026-03-16 15:48:48	BLACKHOLE	Node-42	Critical	Immediate node isolation and route reconfigur...
2026-03-16 15:48:30	BLACKHOLE	Node-42	Critical	Immediate node isolation and route reconfigur...
2026-03-16 15:47:51	BLACKHOLE	Node-1	Critical	Immediate node isolation and route reconfigur...
2026-02-17 18:26:34	BLACKHOLE	Node-1	Critical	Immediate node isolation and route reconfigur...
2026-02-06 14:48:13	BLACKHOLE	Node-1	Critical	Immediate node isolation and route reconfigur...

Fig 7.7: Generated PDF Report of Forensic Attack Logs

The attack logs can also be exported in CSV format for easy data handling and analysis. The CSV file contains details such as attack type, node ID, segment, severity, and mitigation information. This allows users to view, edit, and analyze the data using tools like Excel, making it useful for further study and record keeping.

	B	C	D	E	F	G	H	I	J	K
1	Attack Typ	Node ID	Segment	Severity	Mitigation	Details				
2	Blackhole	Node-1	WSN-Clust	Critical	Immediate	Dataset upload analysis using Unified Ensemble				
3	Grayhole	Node-1	WSN-Clust	High	Isolate no	Dataset upload analysis using Unified Ensemble				
4	Flooding	Node-1	WSN-Clust	Critical	Enable rat	Dataset upload analysis using Unified Ensemble				
5	TDMA	Node-42	WSN-Clust	Medium	Check sch	Dataset upload analysis using Unified Ensemble				
6	Blackhole	Node-42	WSN-Clust	Critical	Immediate	Dataset upload analysis using Unified Ensemble				
7	Blackhole	Node-42	WSN-Clust	Critical	Immediate	Dataset upload analysis using Unified Ensemble				
8	Blackhole	Node-1	WSN-Clust	Critical	Immediate	Dataset upload analysis using Unified Ensemble				
9	Blackhole	Node-1	WSN-Clust	Critical	Immediate	Dataset upload analysis using Unified Ensemble				
10	Blackhole	Node-1	WSN-Clust	Critical	Immediate	Dataset upload analysis using Unified Ensemble				
11	Blackhole	Node-1	WSN-Clust	Critical	Immediate	Dataset upload analysis using Unified Ensemble				
12	Blackhole	Node-1	WSN-Clust	Critical	Immediate	Dataset upload analysis using Unified Ensemble				
13	TDMA	Node-2	WSN-Clust	Medium	Check sch	Dataset upload analysis using Unified Ensemble				
14	Blackhole	Node-2	WSN-Clust	Critical	Immediate	Dataset upload analysis using Unified Ensemble				
15	Normal	Node-1	WSN-Clust	Low	No action	Dataset upload analysis using Unified Ensemble				
16	Normal	Node-910	WSN-Clust	Low	No action	Dataset upload analysis using Unified Ensemble				
17	TDMA	Node-1	WSN-Clust	Medium	Check sch	Dataset upload analysis using Unified Ensemble				

Fig 7.8: CSV Report of Cyberattack Detection Logs

An AI chatbot is integrated into the system to provide quick information about the project and assist users. It can explain project details, features, and functionality in an interactive manner. This improves user experience and helps users understand the system more easily.

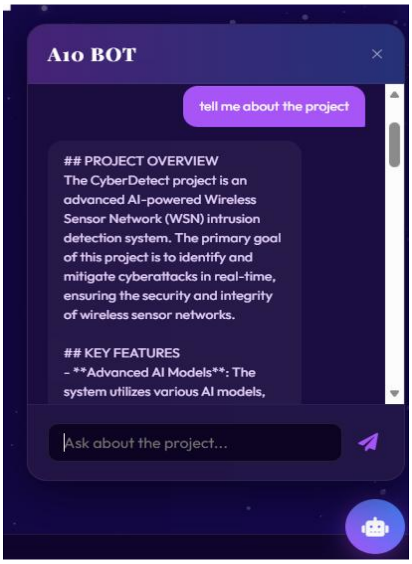


Fig 7.9: AI Chatbot for User Assistance

CHAPTER-8

CONCLUSION AND FUTURE WORK

Conclusion:

In conclusion, the proposed unified ensemble learning based cyberattack detection system for Wireless Sensor Networks has been successfully designed and implemented. The system effectively detects and classifies different types of cyberattacks such as Blackhole, Grayhole, Flooding, and TDMA attacks using a combination of machine learning and deep learning models.

The integration of multiple models in the form of a unified ensemble improves detection accuracy, classification stability, and overall system reliability. The system also supports anomaly detection to identify unknown or emerging attacks, making it more robust in dynamic network environments.

The implementation includes features such as real-time attack detection, dashboard visualization, forensic logging, PDF and CSV report generation, and an AI chatbot for user assistance. These features enhance the usability and practical applicability of the system.

The results demonstrate that the proposed system achieves high accuracy and provides efficient cyberattack detection with reliable performance. Overall, the system can be effectively used in real-world Wireless Sensor Network environments to improve network security and prevent cyber threats.

Future Work:

In future, the system can be extended to support large-scale and real-time network environments. It can be integrated with security systems such as firewalls and monitoring tools for better protection. Further improvements can focus on making the model more adaptive to evolving cyberattack patterns.

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PUBLICATION RELATED TO PROJECT

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Acceptance notification: Registration deadline 4th April 2026

1 message

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Dear Author,
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PROJECT CO-PO MAPPING

Course Outcomes	Program Outcomes											
	1	2	3	4	5	6	7	8	9	10	11	12
Review the literature to understand the current state of art in the field of computer science.	X	X			X							X
Identify a problem and its scope for the proposed work in the selected field of study.	X	X	X									X
Develop appropriate mathematical model or prototype to solve the proposed problem.	X	X	X		X							X
Execute the proposed techniques with team work to solve the problem under consideration.			X	X	X				X	X	X	
Evaluate the proposed techniques with adequate Experimental Analysis.		X		X	X				X	X	X	
Prepare a dissertation following professional ethics.								X	X	X	X	X

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GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

DETAILS OF PROJECT TEAM

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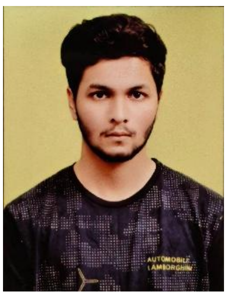	R. Akash 224G1A3206 Computer Science & Engineering (Data Science) Srinivasa Ramanujan Institute of Technology, Anantapur, Andhra Pradesh akashramugalla@gmail.com